# Quiz 4

Graded

**Student**

施明睿

**Total Points**

**16 / 30 pts**

**Question 1**

True or False — **6** / 8 pts

1.1 **a** — **0** / 1 pt

– **0 pts** F

✔ – **1 pt** incorrect

1.2 **b** — **1** / 1 pt

✔ – **0 pts** T

– **1 pt** incorrect

1.3 **c** — **1** / 1 pt

✔ – **0 pts** F

– **1 pt** incorrect

1.4 **d** — **1** / 1 pt

✔ – **0 pts** T

– **1 pt** incorrect

1.5 **e** — **1** / 1 pt

✔ – **0 pts** F

– **1 pt** incorrect

1.6 **f** — **0** / 1 pt

– **0 pts** T

✔ – **1 pt** incorrect

1.7 **g** — **1** / 1 pt

✔ – **0 pts** F

– **1 pt** incorrect

1.8 **h** — **1** / 1 pt

✔ – **0 pts** T

– **1 pt** incorrect

**Question 2**

Fill in the blanks **7** / 13 pts

2.1 **a** **2** / 2 pts

✔ **− 0 pts** 1013

**− 2 pts** incorrect

2.2 **b** **2** / 2 pts

✔ **− 0 pts** 0

**− 2 pts** incorrect

2.3 **c** **0** / 6 pts

**+ 2 pts** log n

**+ 2 pts** 1

**+ 2 pts** n

✔ **+ 0 pts** incorrect

2.4 **d** **3** / 3 pts

✔ **− 0 pts** 10,11,12

**− 1 pt** partially correct (-1 for each missing or wrong)

**− 2 pts** partially correct

**− 3 pts** incorrect

**Question 3**

Heap **3** / 9 pts

3.1 a **1** / 1 pt

✔ **– 0 pts** max-heap

**– 1 pt** incorrect

3.2 b **2** / 3 pts

**– 0 pts** 19-38 (optional since 38 deleted)
19-37
19-36

✔ **– 1 pt** partially correct

**– 2 pts** partially correct

**– 3 pts** incorrect

3.3 c **0** / 2 pts

**– 0 pts** Between 26 and 35

✔ **– 2 pts** incorrect

3.4 d **0** / 3 pts

**– 0 pts** 19,25,26

**– 1 pt** partially correct (-1 for each missing or wrong)

**– 2 pts** partially correct

✔ **– 3 pts** incorrect

CS101A Fall 2025 - Quiz 4
December, 2, 2025 - 20 Minutes

Name: 张明睿 Student ID: 2025533018

**1. (8 points) True or False**

Determine whether the following statements are true or false.

(a) (1') To print the hierarchy of a folder, we should apply BFS on the folder tree. ● True ○ False

(b) (1') If the pre-order traversal and in-order traversal of two binary trees are equal respectively, then the two binary trees are exactly the same. ● True ○ False

(c) (1') Every complete binary tree is also a perfect binary tree. ○ True ● False

(d) (1') A Huffman Coding Tree is a full binary tree. ● True ○ False

(e) (1') Heap sort is a stable sorting algorithm. ○ True ● False

(f) (1') The time complexity of heap sort is $O(n\log(n))$ in both the best-case and the worst-case. ○ True ● False

(g) (1') In a BST, there may be a subtree that is not a BST. ○ True ● False

(h) (1') Suppose there are $n$ nodes in an AVL tree, its minimum possible height is given by $\lceil \log_2(n+1)\rceil - 1$. ● True ○ False

**2. (13 points) Fill in the blanks**

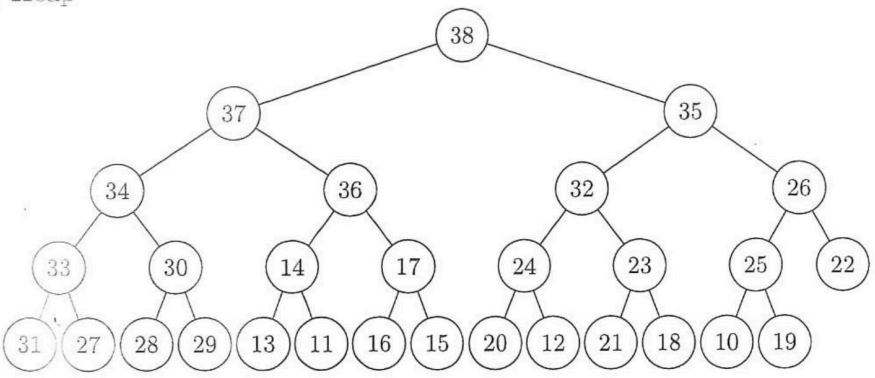
(a) (2') A full binary tree has 2025 nodes. The number of its leaf nodes is 1013.

(b) (2') In a binary tree with n nodes, every node has an odd number of descendants. Then the number of nodes in the tree that have exactly one child is 0.

(c) (6') In a binary max-heap with $n$ elements and duplicated elements are not allowed, the $6^{th}$ largest element can be found in time $O($n$)$ if we can only access the top of the heap. And the $6^{th}$ largest element can be found in time $O($logn$)$ if we can access the array storing the heap. And the $6^{th}$ smallest element can be found in time $O($nlogn$)$ if we can access the array storing the heap. (Give the precise and simplified upper bound)

(d) (3') Elements $[13, 21, 3, 17, 24, 26, 40, 9, 16]$ are inserted sequentially into an empty BST. Then if element $x$ is inserted into the BST as the right child of element 9, then all possible integer values for $x$ could be 10, 11, 12 (all elements are distinct).

**3.** **(9 points)** Heap




(a) (1') Is this heap a max-heap or a min-heap?

max heap

(b) (3') Suppose that you pop the key from the heap above. Write down all the involved swapping operations in order. (for example, $a - b, c - d$ to indicate swapping $a, b$ and then swapping $c, d$)

~~37 - 38~~ 38 - 19 19 - 36

19 - 37 19 - 17

(c) (2') Suppose that you insert a key $k$ into the heap above, and then it is moved to the place of current 26. Write down the possible value range of $k$.

(d) (3') Suppose that inserting the key $x$ was the last operation performed in the binary heap in the figure. That is, after inserting $x$, the heap is shown in the figure above. Write down all possible values of $x$.